

Tech Stack Inventory

Prakhar Jain | Graphic / UI/UX Portfolio Website

Project: prakhar-portfolio (Next.js App Router)

Live: graphic-portfolio-nine-opal.vercel.app

Repo: github.com/prakhar8506/graphic-portfolio

Document purpose: complete inventory of every library, tool, platform service, and concept used to design, build, ship, and operate this site.

Honest scope note: this portfolio is a static content site (TypeScript modules + public assets). It does not run a custom database, user accounts, or REST API. Those rows are marked N/A where they do not apply, with the platform equivalents that still matter (CDN cache, static generation, deployment protection, etc.).

1. CORE RUNTIME & FRAMEWORK

- **Next.js 16.3.3 (App Router)**

React framework: file-based routing, layouts, metadata API, static params, Turbopack in next dev, production builds via next build.

- **React 19.2.8**

UI library - function components, hooks, client/server component split.

- **React DOM 19.2.8**

DOM renderer for React.

- **Node.js (Vercel 24.x runtime)**

Server/build runtime for Next.js on Vercel.

- **TypeScript 5.x**

Typed JavaScript across app/, components/, lib/. Strict mode in tsconfig.

- **ECMAScript / ES2017+ target**

Compiled/bundled for modern browsers.

2. APPLICATION LIBRARIES (NPM)

Runtime dependencies

- **GSAP 3.15+**

GreenSock Animation Platform - loader timeline (fade-in wordmark, 000-100 counter, slide-away panel). Timelines, fromTo, power easings, kill on unmount.

- **Lenis 1.3+**

Smooth scrolling engine - wheel smoothing, start/stop with menu open, requestAnimationFrame loop, Lenis CSS import.

Styling

- **Tailwind CSS 4**

Utility + design tokens via @import "tailwindcss" and @theme inline in globals.css.

- **@tailwindcss/postcss**

PostCSS plugin bridging Tailwind 4 into the CSS pipeline.

- **PostCSS**

CSS transform pipeline (postcss.config.mjs).

Dev / quality

- **ESLint 9**

Lint runner (npm run lint).

- **eslint-config-next 16.3.3**

Next.js Core Web Vitals + TypeScript ESLint presets.

- **@types/node, @types/react, @types/react-dom**

TypeScript type definitions.

- **npm + package-lock.json**

Package manager and lockfile for reproducible installs.

3. NEXT.JS FEATURES & APIS USED

- **App Router (app/)**

Routes: /, /works, /works/[slug], not-found.

- **Layouts (app/layout.tsx)**

Shared shell: fonts, loader, nav, menu, grain, providers.

- **Server Components (default)**

Most pages/sections render on the server with no client JS.

- **Client Components ("use client")**

Hero, Loader, Nav, MenuOverlay, WorksPreview, SmoothScroll, SiteProvider, Cursor - for hooks, GSAP, Lenis, hover state.

- **next/link**

Client-side navigation between pages and hash sections.

- **next/navigation**

usePathname, notFound().

- **Metadata API**

export const metadata and generateMetadata for titles/descriptions.

- **generateStaticParams**

Prebuilds all /works/[slug] case-study pages at build time.

- **next/font/google**

IBM Plex Mono self-hosted/optimized font loading.

- **Path aliases (@/*)**

tsconfig paths for clean imports.

- **next.config.ts**

devIndicators: false - hide Next.js N badge in local UI.

- **Turbopack**

Default bundler for next dev (HMR).

- **create-next-app**

Initial project scaffold (TypeScript, Tailwind, ESLint, App Router).

4. REACT PATTERNS & BROWSER APIS

- **Hooks**

useState, useEffect, useRef, useCallback, useMemo, useContext, createContext.

- **React Context (SiteProvider)**

Global UI state: menuOpen, loaded - shared by Loader, Nav, SmoothScroll, MenuOverlay.

- **requestAnimationFrame**

Lenis smooth-scroll loop.

- **matchMedia / prefers-reduced-motion**

Skip loader animation when user prefers reduced motion.

- **DOM APIs**

classList, style overflow lock, transforms for hover preview window.

- **Timers**

setTimeout fail-safe on loader; clearTimeout on cleanup.

- **Mouse events**

onMouseMove / onMouseEnter / onMouseLeave for project preview window.

5. CSS, TYPOGRAPHY & VISUAL SYSTEM

- **Custom CSS architecture (globals.css)**
Design tokens as CSS variables (--bg, --fg, --muted, --pad, type scale, easings).
- **Neue Montreal (Fontshare CDN)**
Primary display/sans via api.fontshare.com.
- **IBM Plex Mono (Google Fonts + next/font)**
Labels / mono accents.
- **CSS animations / keyframes**
Hero wordmark fade-up; tools marquee.
- **mix-blend-mode**
Nav difference blend; grain multiply.
- **object-fit: contain**
Full project imagery without crop slits.
- **Sticky / fixed / absolute positioning**
Nav, loader overlay, hero corners, hover work-window.
- **clamp() fluid type & spacing**
Responsive type without breakpoint sprawl.
- **SVG noise grain overlay**
Inline data-URI feTurbulence grain.
- **CSS transitions**
Underline hover, menu overlay slide, work-window opacity.
- **Media queries**
Mobile layout for works list, footer, hero.
- **::selection styling**
Inverted black/white selection.
- **Accessibility CSS**
sr-only skip link; focus:not-sr-only.

6. CONTENT ARCHITECTURE (NO DATABASE)

- **TypeScript content modules**
lib/site.ts - identity, socials, live sites, experience, skills, tools.
- **lib/works.ts**
Typed Work[] catalog: slug, title, tag, year, image, gallery, href, services, body.
- **Static assets (public/works/)**
JPG covers and gallery frames served as static files.
- **Database**
N/A - no Postgres/MySQL/Mongo/Supabase. Content is compile-time data + static files.
- **ORM / query layer**
N/A - plain TypeScript arrays and helpers (getWork, workHost).
- **CMS**
N/A - content edited in source (git-backed content).

System-design equivalent: content is a build-time data store. Changing a project = edit TS + commit + redeploy.

7. AUTH, SECURITY & ACCESS CONTROL

- **Application user auth**
N/A - no login, sessions, JWT, Clerk, Auth.js, or OAuth for site visitors.
- **Contact actions**
mailto: and tel: links (email / phone deep links) - no form backend.
- **Vercel Authentication (Deployment Protection)**
Platform SSO gate on .vercel.app deploy URLs; disabled for public portfolio access. Production alias is public.

- **Vercel Deployment Protection settings**
ssoProtection off; password protection off; trusted IPs off.
- **HTTPS / TLS**
Terminated by Vercel for all deployments.
- **Security headers (platform)**
Strict-Transport-Security and related defaults from Vercel.
- **rel="noreferrer" on external links**
Referrer hygiene for outbound Instagram, Drive, live sites.
- **aria-* / semantic HTML**
aria-label, aria-expanded, aria-hidden on loader/overlays; region labels.

8. CACHING, CDN & PERFORMANCE

- **Vercel Edge CDN**
Global static/asset delivery for HTML, JS, CSS, images.
- **Next.js static generation / prerender**
Pages prerendered at build (x-nextjs-prerender / Vercel cache headers in production).
- **Browser HTTP cache**
Cache-Control from platform (must-revalidate on HTML; long-lived hashed assets).
- **Redis / KV / application cache**
N/A - no Upstash, no custom cache layer.
- **Image presentation**
Plain tags for portfolio frames (object-fit: contain); not next/image in current implementation.
- **Font optimization**
next/font for IBM Plex Mono; Fontshare/Google CSS for Neue Montreal.
- **Code splitting / route chunks**
Automatic via Next.js App Router bundling.
- **Turbopack HMR**
Fast local reload during development.

9. APIS & DATA FETCHING

- **Custom REST/GraphQL API routes**
N/A - no app/api/ handlers.
- **Server Actions**
N/A - no form mutations.
- **External live-site links**
Outbound HTTPS to Sarveio, Carrystyle, Sanmati Abhushan (marketing sites, not this app's backend).
- **Font CDNs**
Fontshare API CSS; Google Fonts CSS.
- **Vercel REST / project APIs (ops)**
Used via Vercel dashboard/MCP to manage deployment protection, projects, domains.
- **GitHub API (ops)**
Repo listing, deploy status URLs for linked projects.

System-design equivalent: this site is a static brochure + case-study reader. Data flow is Build -> Static assets -> CDN -> Browser.

10. SYSTEM DESIGN & FRONTEND ARCHITECTURE CONCEPTS

- **Component composition**
Page = Hero + Manifesto + WorksPreview + StudioVideo + Studio + Footer.
- **Separation of concerns**

lib/ (data) | components/ (UI) | app/ (routes) | public/ (assets).

- **Client/server boundary**
Interactive islands ("use client") inside mostly server-rendered tree.
- **Provider pattern**
SiteProvider for cross-cutting UI state.
- **Typed domain model**
Work type with optional href/gallery; helpers for host labels.
- **Static site generation (SSG)**
Case studies from generateStaticParams.
- **Long-scroll marketing layout**
Home as one scroll narrative + deep links to /works/[slug].
- **Progressive enhancement / a11y**
Skip link, reduced-motion path, semantic landmarks.
- **Hover preview UX**
Cursor-following project window -> click to detail route.
- **Loader gate**
Blocks Lenis until intro completes; fail-safe timeout.
- **Responsive design**
Desktop-first typographic composition with mobile grid collapse.
- **Design tokens**
CSS variables for color, type, easing - single visual language.
- **CDN + immutable deploys**
Each Vercel deploy is atomic; aliases point to production.
- **GitOps content workflow**
Source-of-truth in git; push to main -> Vercel rebuild.

11. HOSTING, DOMAINS & ENVIRONMENTS

- **Vercel**
Hosting, builds, previews, production aliases, CDN, HTTPS.
- **Production domain**
graphic-portfolio-nine-opal.vercel.app (and team aliases).
- **Preview / deploy URLs**
Per-deployment *.vercel.app hostnames (often SSO-protected by default).
- **Git integration**
GitHub repo prakhar8506/graphic-portfolio linked to Vercel project graphic-portfolio.
- **Environments**
Local next dev (e.g. :3004) | Preview | Production.
- **npm scripts**
dev, build, start, lint.

12. VERSION CONTROL, GITHUB & AUTOMATIONS

- **Git**
Source control - commits, branches, working tree.
- **GitHub**
Remote hosting for graphic-portfolio (and related repos: sanmatiWeb, carrystyle-bags).
- **Vercel GitHub automation**
Push to main -> automatic production deploy; PR/branch -> preview deploy.
- **GitHub Deployments / commit statuses**
Vercel posts deployment status + inspector URLs on commits.

- **CI concept (platform-managed)**
Build + lint surface via Next/ESLint locally; deploy pipeline owned by Vercel Git integration.
- **.gitignore**
Excludes node_modules, .next, env secrets, etc.
- **Lockfile discipline**
package-lock.json committed for reproducible builds.

13. DEVELOPER TOOLS & AI ASSIST

- **Cursor IDE**
Primary editor / agent-assisted coding.
- **Claude / ChatGPT**
Copy, resume, architecture and debugging assistance (listed in site tools).
- **Terminal / zsh**
Local shell for npm, curl, git.
- **Browser DevTools**
Layout, network, console verification.
- **Playwright / browser automation (dev)**
Used during build/QA to verify hover, routes, public HTTP 200s.
- **curl**
Smoke-check production headers and SSO redirects.
- **Vercel CLI / MCP / dashboard**
Project linking, deployment protection, logs, domains.

14. DESIGN, MOTION & ASSET PRODUCTION TOOLS

Used to create the work shown on the site (and listed in the Studio tools section), even when not imported as npm packages.

Craft

- **Figma**
UI/UX, layout, product screens.
- **Adobe Photoshop**
Image edit, thumbnails, composites.
- **Adobe Illustrator**
Vector / brand marks.

Motion / video

- **DaVinci Resolve**
Edit + colour.
- **Adobe After Effects**
Motion cues / motion graphics.
- **Adobe Premiere Pro**
Video editing.

AI image / product assist

- **Midjourney, Runway, Flux, v0**
Visual exploration / generation / UI scaffolding (practice toolkit).

Web builders (related practice)

- **Framer, Spline, Webflow**
Listed in site tools - spatial/web prototyping adjacent to this Next build.

15. OPERATING SYSTEM & LOCAL STACK

- **macOS (darwin)**
Local development machine.
- **Homebrew / Node toolchain**
Node + npm environment.
- **File system static serving**
Next serves public/ at URL root.

16. LINKED LIVE PROJECTS (OUTBOUND)

- **Sarveio - sarveio.in**
Restaurant OS / AR dining marketing site.
- **Carrystyle - carrystyle-bags.vercel.app**
Bag manufacturing brand site.
- **Sanmati Abhushan - sanmati-web.vercel.app**
Jewellery brand site (use production alias, not hashed deploy URLs).
- **Google Drive portfolio folder**
Shared work archive linked from footer/hero.
- **Instagram @uiux.prakhar**
Social proof / process.

17. QUICK CHECKLIST - EVERYTHING TOUCHED

Next.js 16 App Router | React 19 | React DOM | TypeScript | Node.js | GSAP | Lenis | Tailwind CSS 4 | PostCSS | @tailwindcss/postcss | ESLint + eslint-config-next | npm / package-lock | next/link | next/navigation | next/font | Metadata API | generateStaticParams | Server Components | Client Components | React Context | CSS variables / custom CSS | Fontshare Neue Montreal | IBM Plex Mono | requestAnimationFrame | matchMedia (reduced motion) | Static content modules (lib/) | public/ static images | Git | GitHub | Vercel (host + CDN + Git auto-deploy) | HTTPS/TLS | Deployment Protection / Vercel Auth (ops) | CDN caching / prerender | mailto: / tel: | Cursor | Claude / ChatGPT | Figma | Photoshop | Illustrator | DaVinci Resolve | After Effects | Premiere Pro | Midjourney / Runway / Flux / v0 | Framer / Spline / Webflow (practice) | curl / browser QA | Accessibility (skip link, ARIA, reduced motion)

18. EXPLICITLY NOT PART OF THIS CODEBASE

- **Database**
No SQL/NoSQL, no Prisma/Drizzle, no Supabase/Neon.
- **App-level Auth**
No user accounts, sessions, cookies for visitors.
- **Custom Backend API**
No Route Handlers / tRPC / GraphQL server.
- **Application Redis/KV cache**
Relies on Vercel/Next static + CDN only.
- **Analytics SDK in code**
Not wired in source (platform analytics optional via Vercel dashboard).
- **Payment / e-commerce backend**
Case studies only; commerce lives on linked brand sites.

If you later add a CMS, contact form, or dashboard, typical stack additions would be: Postgres (Neon) + Prisma, Server Actions or Route Handlers, Resend/email API, Auth.js or Clerk, and Upstash Redis for rate limits - none of those ship in the current portfolio.

End of inventory

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